

### 3.1.3 Developments in new materials

Students should be aware of developments in new materials.

#### Modern materials

| Content                                                                                                                                                                                                                             | Potential links to maths and science                                                                                                                                    |
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| Developments made through the invention of new or improved processes eg Graphene, Metal foams and Titanium.<br><br>Alterations to perform a particular function eg Coated metals, Liquid Crystal Displays (LCDs) and Nanomaterials. | Classification of the types of properties of a range of materials.<br><br>Selecting appropriate materials.<br><br>Extracting information from technical specifications. |

#### Smart materials

| Content                                                                                                                                                                                                                                             | Potential links to maths and science                                                                                                                                    |
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| That materials can have one or more properties that can be significantly changed in a controlled fashion by external stimuli, such as stress, temperature, moisture, or PH eg shape memory alloys, thermochromic pigments and photochromic pigments | Classification of the types of properties of a range of materials.<br><br>Selecting appropriate materials.<br><br>Extracting information from technical specifications. |

#### Composite materials

| Content                                                                                                                                                                                       | Potential links to maths and science                                                                                                                                    |
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| That composite materials are produced by combining two or more different materials to create an enhanced material eg glass reinforced plastic (GRP) and carbonfibre reinforced plastic (CRP). | Classification of the types of properties of a range of materials.<br><br>Selecting appropriate materials.<br><br>Extracting information from technical specifications. |

## Technical textiles

| Content                                                                                                                                                  | Potential links to maths and science                                                                                                                            |
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| How fibres can be spun to make enhanced fabrics eg conductive fabrics, fire resistant fabrics, kevlar and microfibres incorporating micro encapsulation. | Classification of the types of properties of a range of materials.<br>Selecting appropriate materials.<br>Extracting information from technical specifications. |